Towards Enhanced Reasoning Capabilities of LLMs

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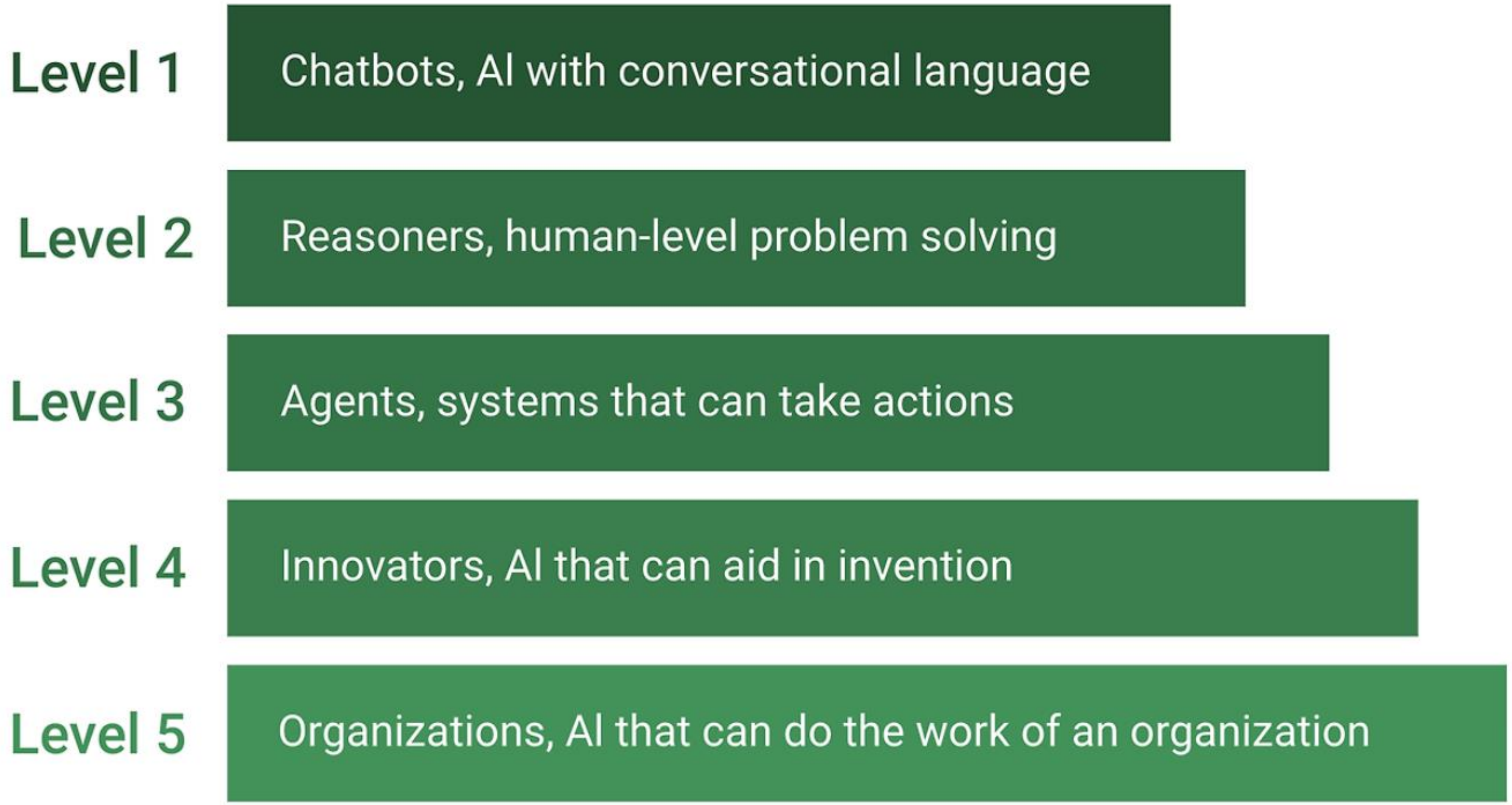
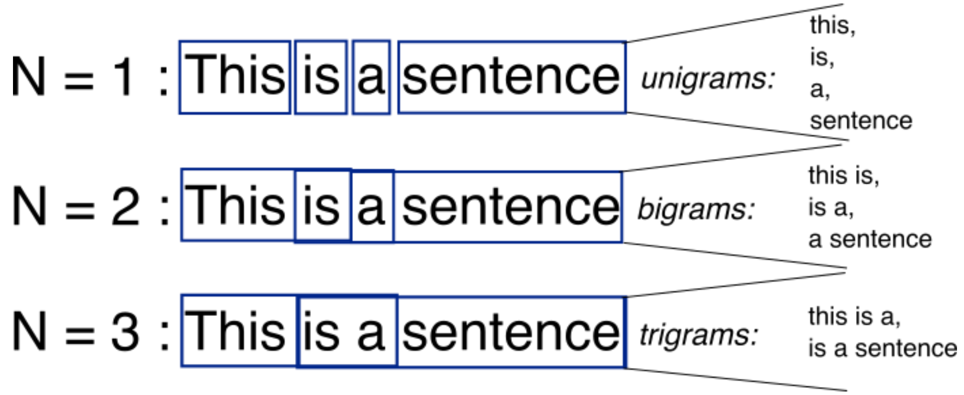
OpenAI’s 5 Steps to AGI



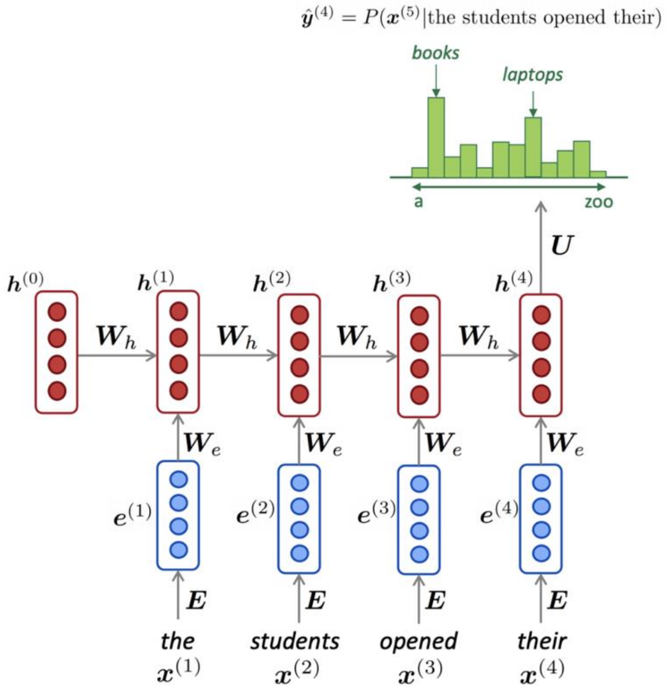
Language Model (LM)

- Language models (LMs) perform the task of assigning probabilities to word sequences
  - It models **the fluency** and **plausibility** of word sequence (i.e., texts)
  - $P(X) = P(x_1, x_2, \dots, x_T) = \prod_{t=1} P(x_t | x_1, x_2, \dots, x_{t-1})$
  - P("I love you") $>$ P("You love I")
- Early LMs rely on statistical approach based on counts of word occurrences
  - N-gram LM: It uses a fixed (i.e., N-1) number of previous words to predict the next word

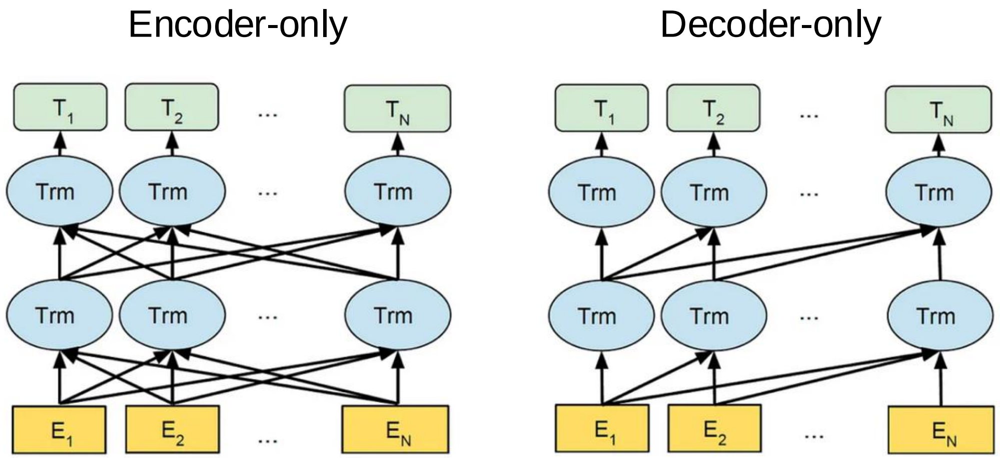


Language Model (LM)

- **Deep neural networks (DNNs)**, with their ability to capture intricate patterns, revolutionized language modeling (e.g., Recurrent neural networks, Transformers)
  - Due to the complexity, more nuanced approaches beyond n-grams is required



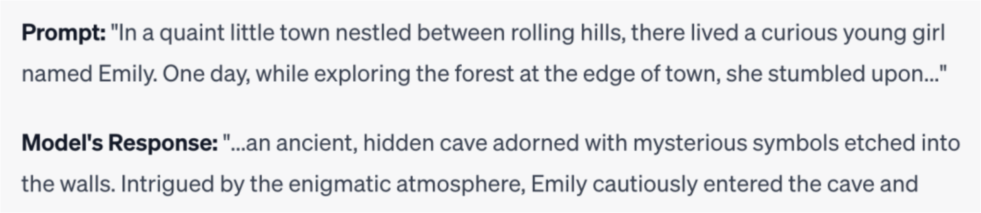
RNN-based language model



Transformer-based language model

Power of Large Language Model

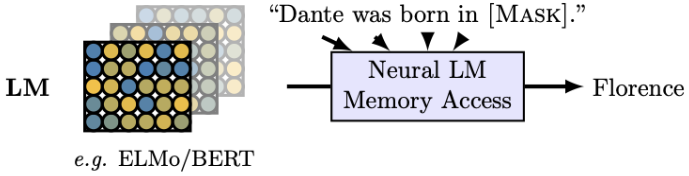
- Language modeling
  - It can generate **plausible texts** by predicting the next words or phrases
  - e.g., masked word prediction, open-ended text generation
- Knowledge base
  - LMs learn the **factual knowledge** encoded in the text data they are trained on
  - e.g., knowledge probing
- Reasoning capability
  - Scaling up the size of LMs (→ LLM) has shown to confer a range of **reasoning abilities**



Prompt: "In a quaint little town nestled between rolling hills, there lived a curious young girl named Emily. One day, while exploring the forest at the edge of town, she stumbled upon..."

Model's Response: "...an ancient, hidden cave adorned with mysterious symbols etched into the walls. Intrigued by the enigmatic atmosphere, Emily cautiously entered the cave and

Open-ended text generation



Knowledge probing

Open source vs. Proprietary LLM

- Open source LLM
  - **Publicly available**, can be modified and distributed freely
  - Users can modify the model for specific needs or tasks
  - Typically free to use, may involve costs for customization or support
  - e.g., Llama, Gemma, DeepSeek, Qwen, OLMo, …
- Proprietary LLM
  - Developed by a company with **restricted access** to the source code
  - Limited customization options, controlled by the owning company
  - Usually involves licensing fees, costs may vary based on usage
  - e.g., GPT, o1, Gemini, Claude, …

LM Prompting

- **LM prompting** refers to the practice of providing specific instructions, questions, or cues as input to a LM in order to guide its generation of text
  - Prompt is a piece of text or an input given by a user to instruct the model and receive a desired response
  - The effectiveness and accuracy of LMs depend on well-crafted prompts that provide clarity, specificity, and sufficient context
- **Prompt Engineering** focuses on how to communicate with LM to steer its behavior for desired outcomes without updating the model weights
  - Prompt engineer is shaping up to be a promising AI-related career

LM Prompting

- Example of ChatGPT (GPT-3.5)

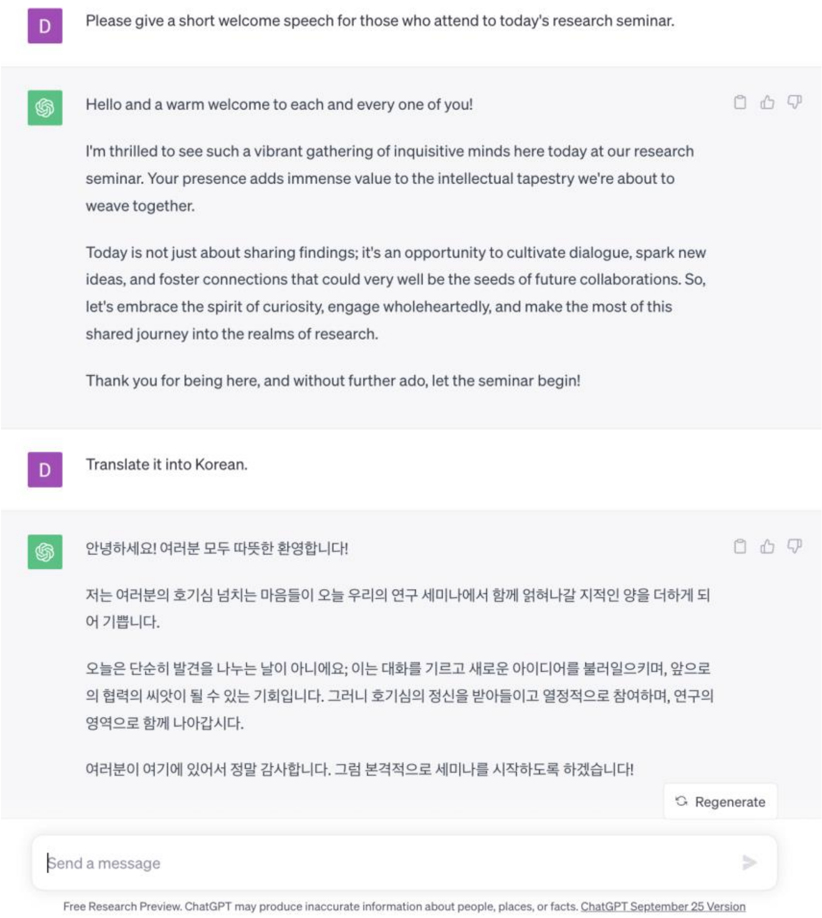
Prompt →

Model Output →

Prompt →

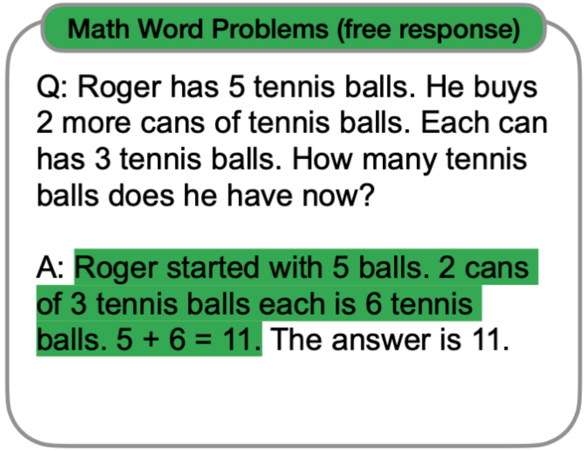
Model Output →

Prompt →

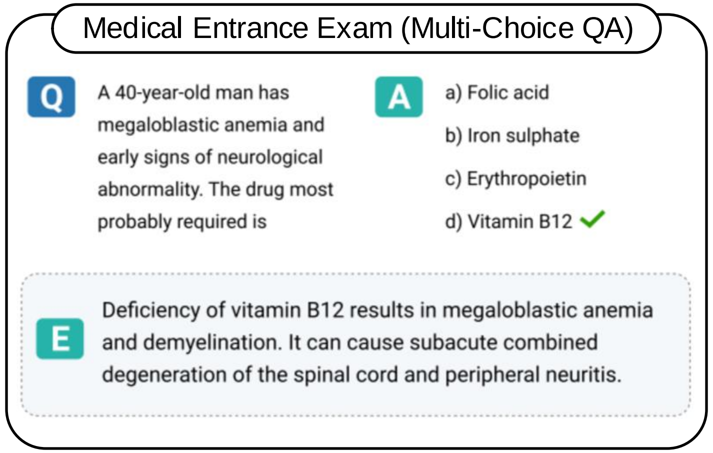


Challenging Tasks in Reasoning

- Arithmetic (or mathematical) reasoning
  - The ability to perform reasoning on math word problems (MWP)
- Domain-specialized reasoning
  - The ability to perform reasoning within a specific knowledge domain (e.g., *medicine*, *law*, or *finance*), using domain-specific facts, principles, and contextual understanding



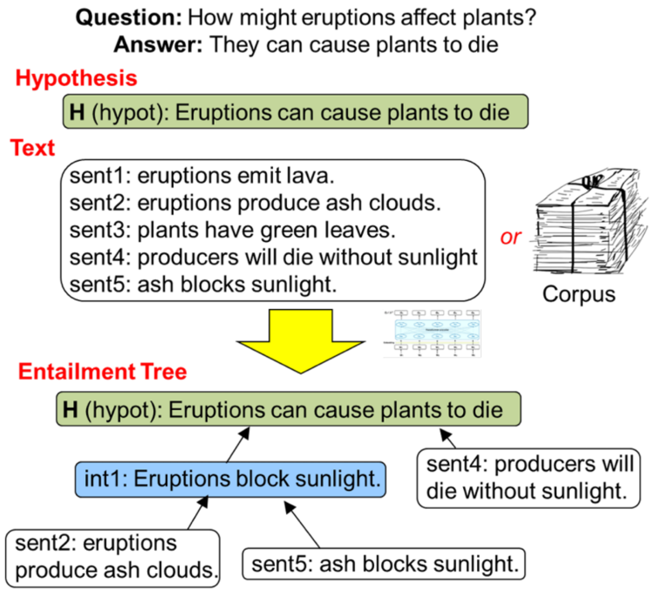
Arithmetic reasoning



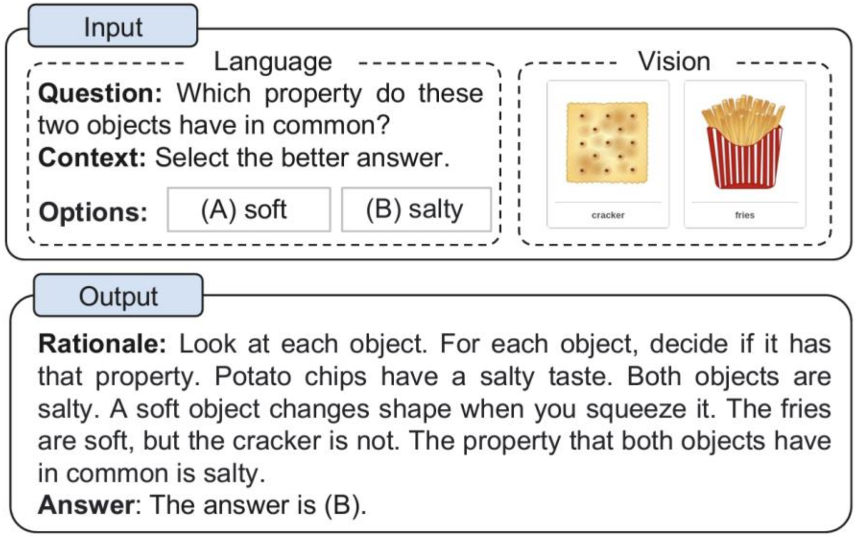
Domain-specialized reasoning

Challenging Tasks in Reasoning

- Logical reasoning (→ inductive, deductive, abductive)
  - The ability to systematically draw conclusions based on logical principles
- Multimodal reasoning
  - The ability to reason by utilizing information across different modalities (e.g., text + image)



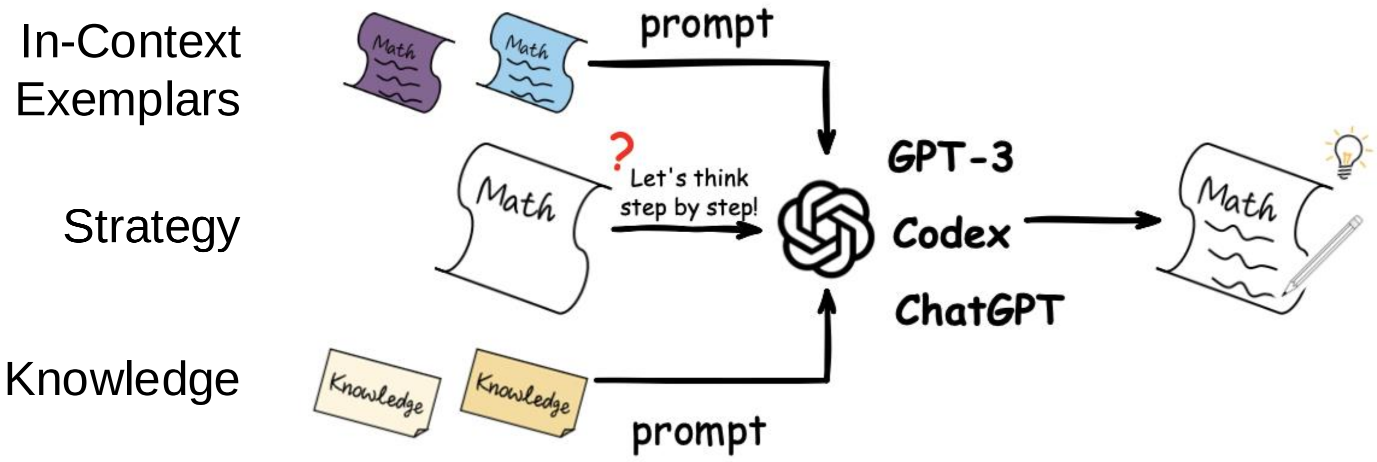
Logical reasoning



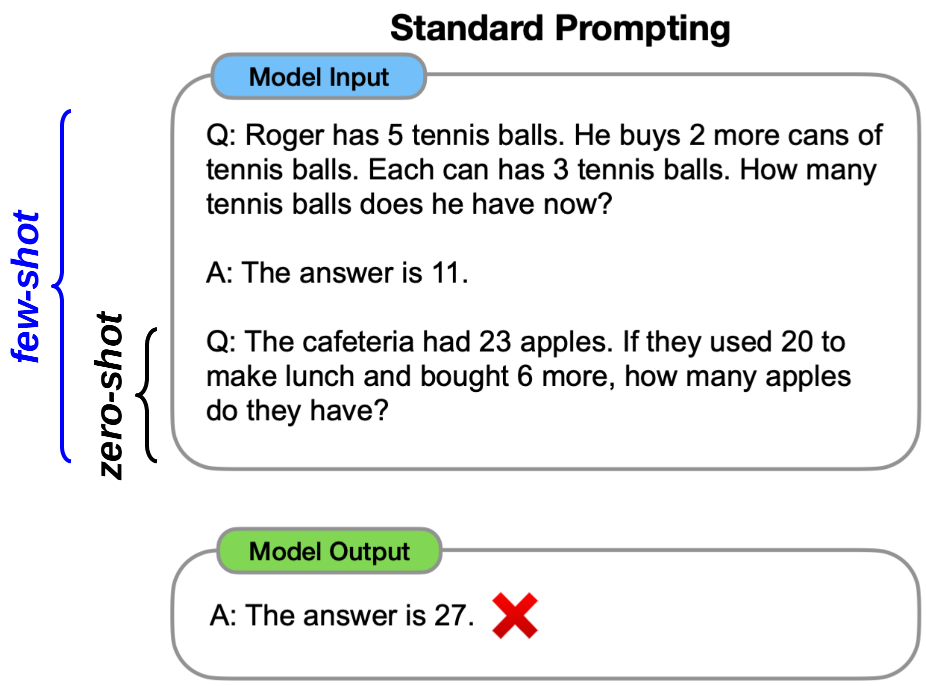
Multimodal reasoning

Reasoning with LM Prompting

- Various ***prompting strategies*** can unlock reasoning abilities
  - e.g., chain-of-thought (CoT) prompting, generated knowledge prompting
  - This can dramatically narrow the gap between human and machine intelligence



Standard LM Prompting



- Given the reasoning question $\mathcal{Q}$, prompt $\mathcal{T}$, and parameterized probabilistic model p_{LM},
  - We aim to maximize the likelihood of answer $\mathcal{A}$

$$p(\mathcal{A} \mid \mathcal{T}, \mathcal{Q}) = \prod_{i=1}^{|\mathcal{A}|} p_{\mathrm{LM}}\left(a_i \mid \mathcal{T}, \mathcal{Q}, a_{<i}\right)$$

  - where a_i and $|\mathcal{A}|$ denotes the i-th token
- For ***few-shot prompting***, $\mathcal{T}$ is comprised of exemplars of $(\mathcal{Q}, \mathcal{A})$ pair

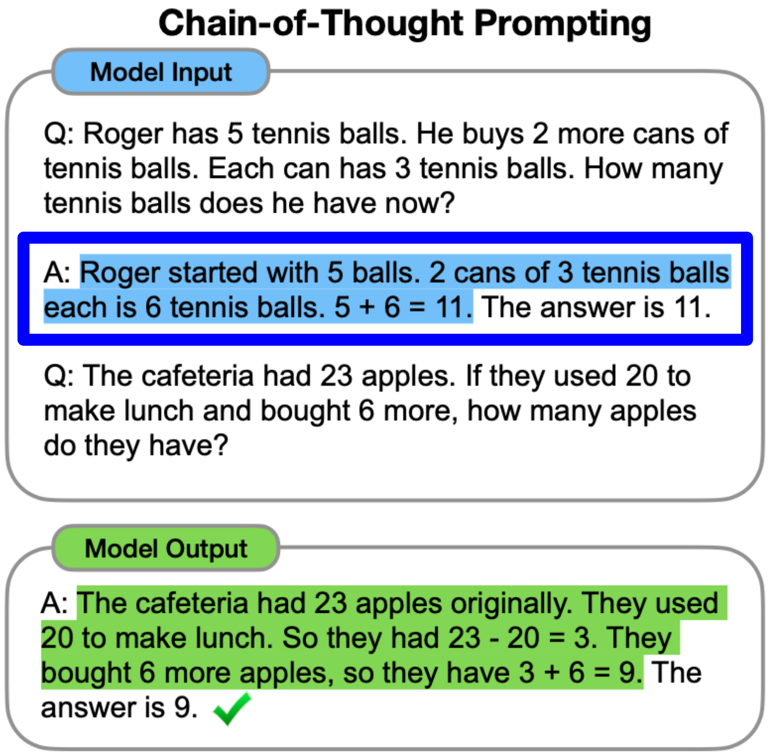
1) Wei et al., Chain-of-Thought Prompting Elicits Reasoning in Large Language Models, NeurIPS 2022

Taxonomy of Methods

- **Strategy-enhanced** reasoning
  - It focuses on optimizing the reasoning strategy with prompting
    - Prompt engineering
    - Process optimization
    - External engine
- **Knowledge-enhanced** reasoning
  - It focuses on knowledge enhancement with prompting
    - Explicit knowledge
    - Implicit knowledge

Strategy-Enhanced Reasoning

- Prompt Engineering 1 – **Single-stage methods** (Chain-of-Thought prompting)

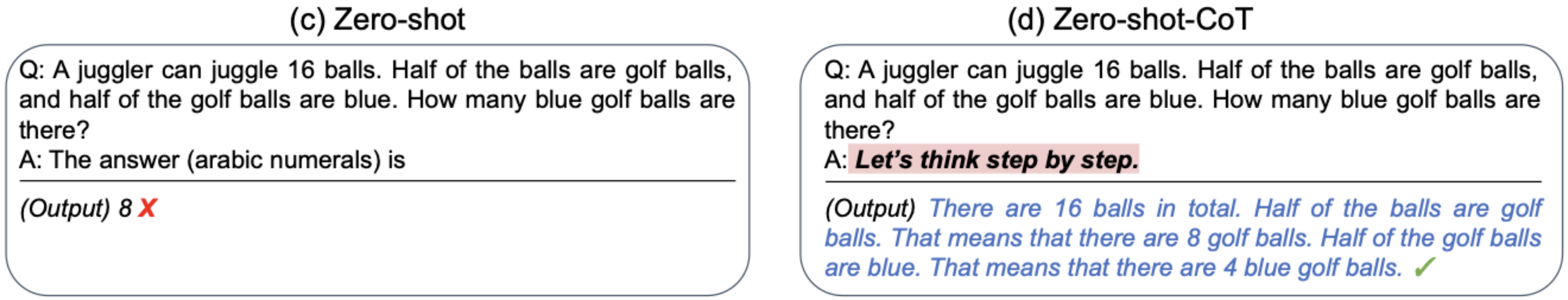


- **Chain-of-Thought (CoT) prompting** adds **a series of intermediate reasoning steps** into exemplars of few-shot prompt to induce large LMs **to generate a reasoning process** before answering
  - It significantly improves the ability of large language models to perform complex reasoning
  - It particularly improves performance on a range of arithmetic, commonsense, and symbolic reasoning

1) Wei et al., Chain-of-Thought Prompting Elicits Reasoning in Large Language Models, NeurIPS 2022

Strategy-Enhanced Reasoning

- Prompt Engineering 1 – **Single-stage methods** (Chain-of-Thought prompting)

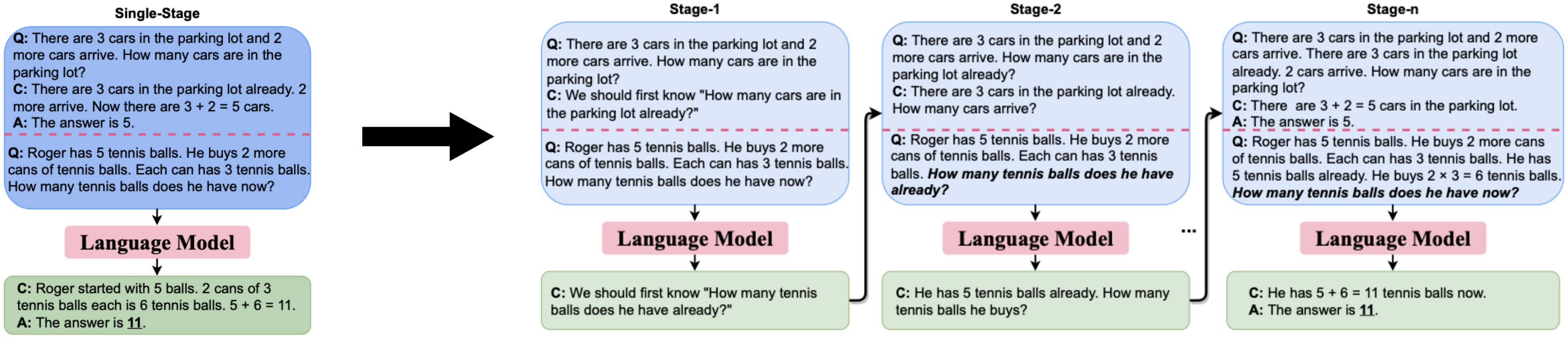


  - LMs are also **zero-shot reasoners** without needing extra exemplars
  - By concatenating "*Let's think step by step!*", LMs can consciously generate reasoning steps

1) Kojima et al., Large language models are zero-shot reasoners, NeurIPS 2022

Strategy-Enhanced Reasoning

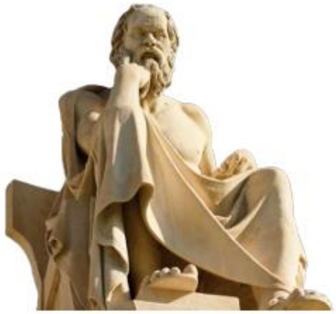
- Prompt Engineering 2 – **Multi-stage methods**

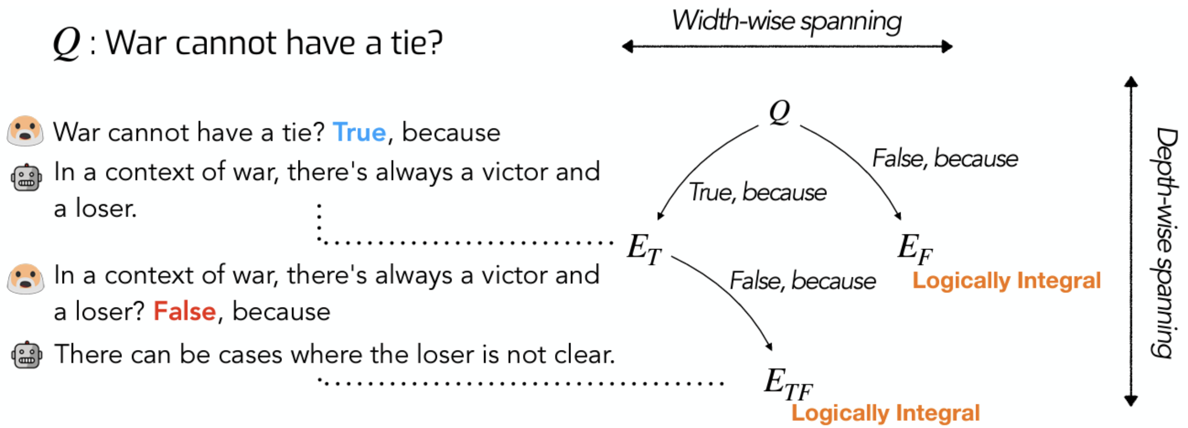


- It is challenging for humans to come up with the whole reasoning process in one stroke
- A more intuitive solution is **to decompose** a complex problem into simpler ones and **to reason stage by stage**
  - It transforms one-stage prompting into multi-stage prompting
  - once input-output → multi-times of input-output

Strategy-Enhanced Reasoning

- Prompt Engineering 2 – **Multi-stage methods**



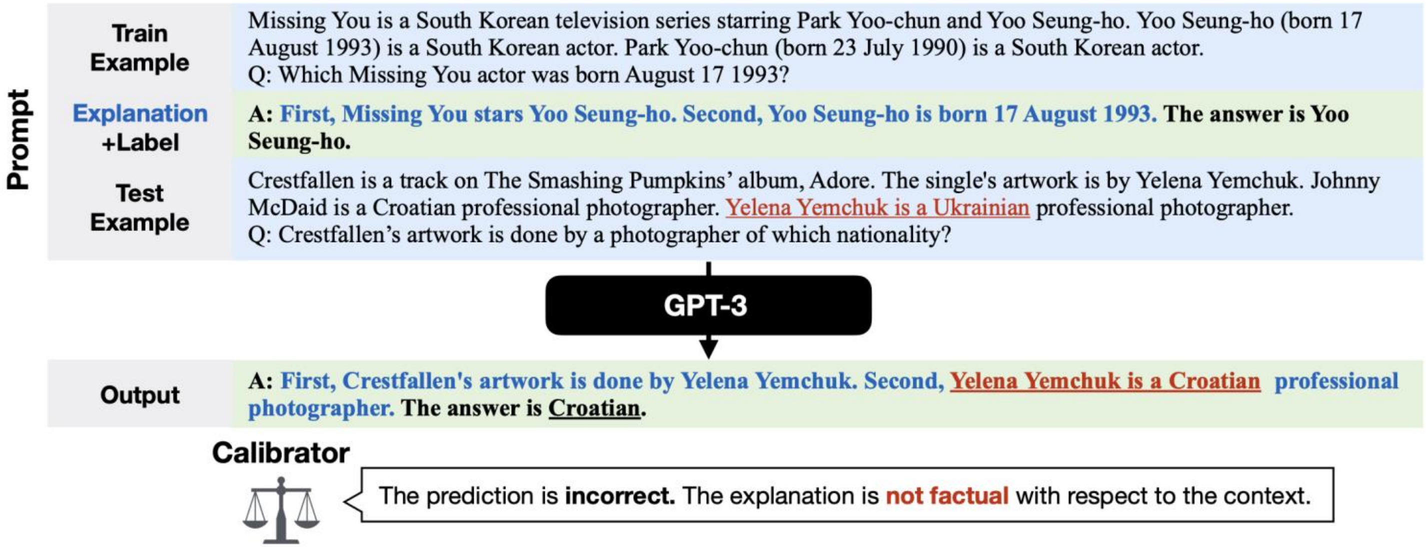


- **Maieutic prompting** aims to infer a correct answer to a question even from the unreliable generations of LM
  - Maieutic tree generation – abductive explanation for “True” and “False”
  - Logical inference between explanations

1) Jung et al., Maieutic Prompting: Logically Consistent Reasoning with Recursive Explanations, EMNLP 2022

Strategy-Enhanced Reasoning

- Process Optimization 1 – **Self-Optimization**

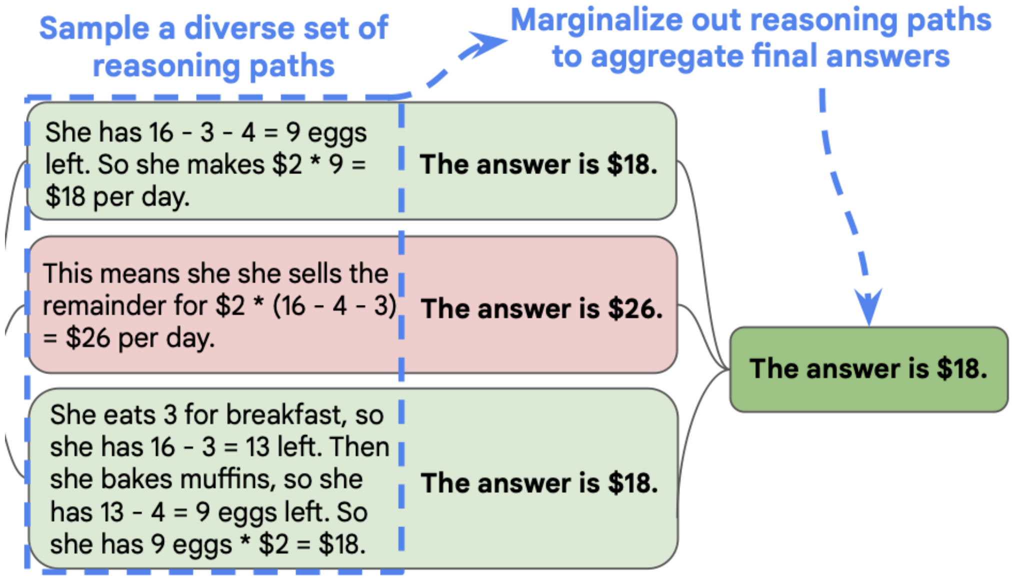


- Self-optimization applies **an optimizer module** to calibrate a single reasoning process
- To mitigate the influence of the unreliability of rationales, a calibrator is utilized to tune the probabilities of a prediction based on the score which reflects the factuality of a rationale

1) Ye and Durrett, The Unreliability of Explanations in Few-shot Prompting for Textual Reasoning, NeurIPS 2022

Strategy-Enhanced Reasoning

- Process Optimization 2 – **Ensemble-Optimization**

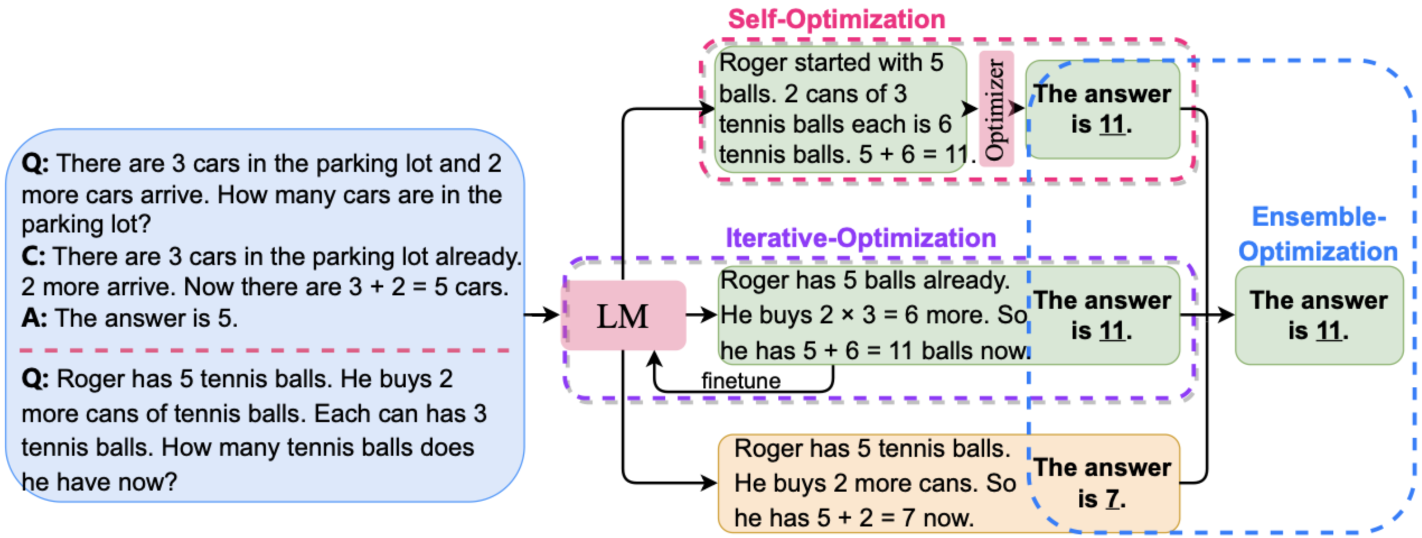


  - Ensemble-optimization assembles multiple reasoning processes to calibrate the answer
  - It obtains multiple reasoning paths and generates the most consistent answer by majority vote

1) Wang et al., Self-Consistency Improves Chain of Thought Reasoning in Language Models, ICLR 2023

Strategy-Enhanced Reasoning

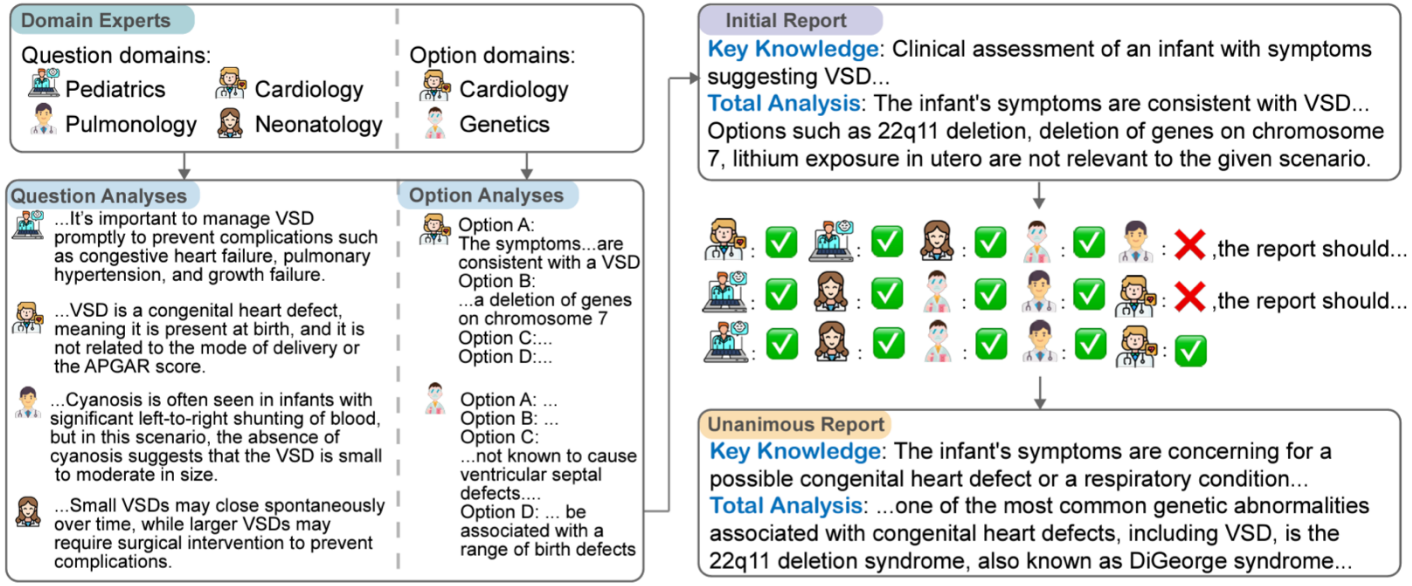
- Process Optimization 3 – **Iterative-Optimization**



- Iterative-optimization calibrates reasoning processes by iteratively fine-tuning the LM
- It generates multiple reasoning paths and fine-tunes the most consistent self-generated answers

Strategy-Enhanced Reasoning

- Process Optimization 4 – **Collaborative-Optimization**

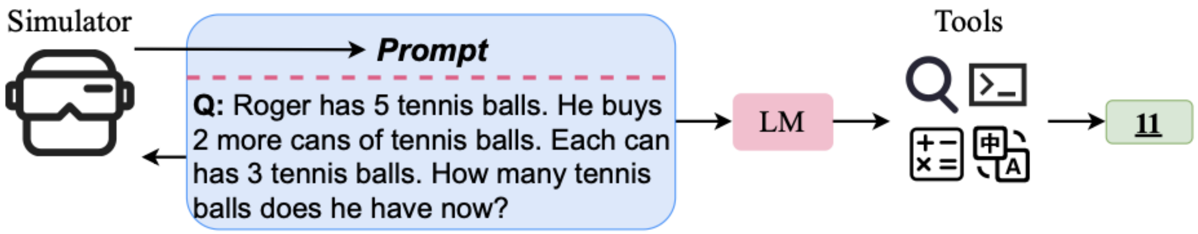


- Collaborative-optimization leverages agents in a role-playing setting
- Multiple agents participate in a collaborative multi-round discussion, thereby enhancing LLM proficiency and reasoning capabilities

1) Tang et al., MedAgents: Large Language Models as Collaborators for Zero-shot Medical Reasoning, ACL 2024

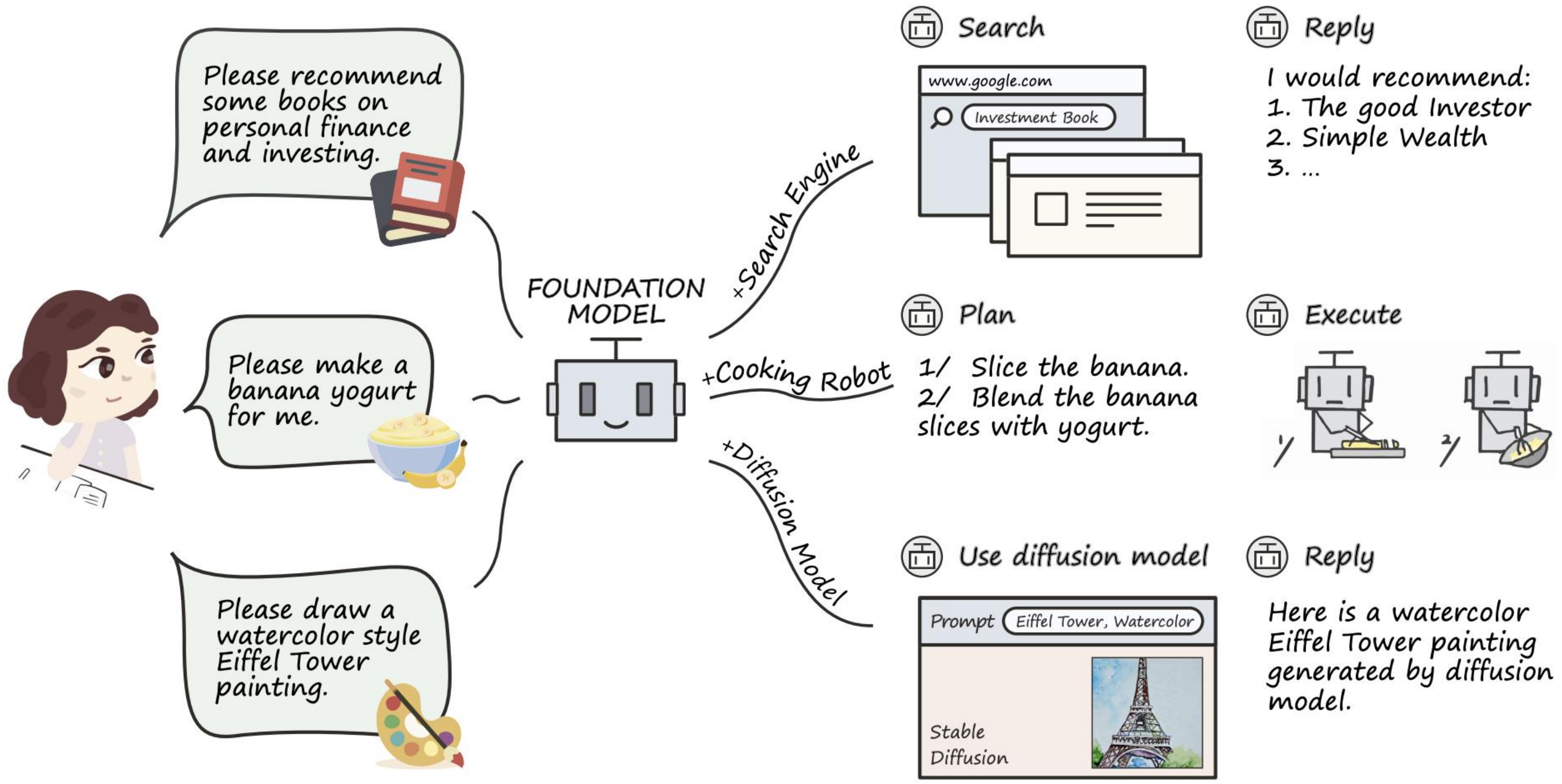
Strategy-Enhanced Reasoning

- External Engine
    - **Physical-Optimization**: Given a physical reasoning question, it utilizes a computational physics engine to simulate the physical process.
    - **Code Interpreter**: Programs yield advantage behaviors in robustness and interpretability and can better illustrate complex structures and deduct complex calculations.
    - **Tool Learning**: The models are trained by integrating the usage of various tools, including calculators, Q&A systems, search engines, etc.



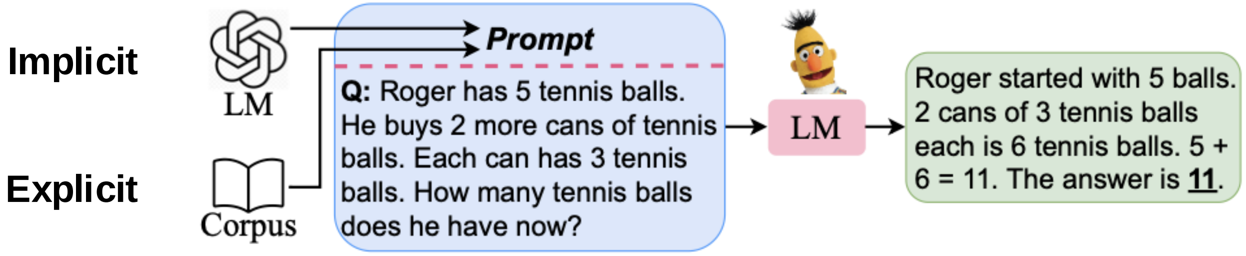
Strategy-Enhanced Reasoning

- External Engine



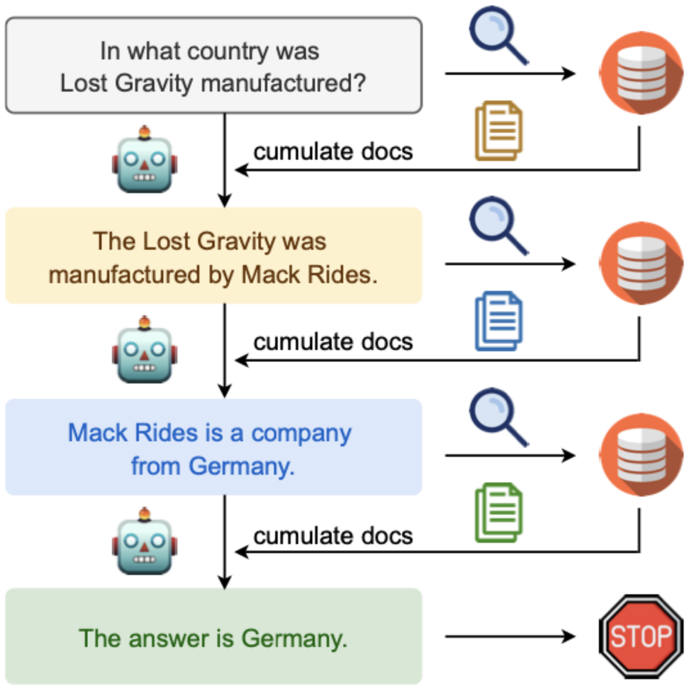
Knowledge-Enhanced Reasoning

- Explicit (or *External, Symbolic*) Knowledge
  - Knowledge that can be easily articulated, codified, and shared with others
  - Consciously processed and can be precisely conveyed, documented, and transferred
    - e.g., Mathematical Equations, Grammar Rule
- Implicit (or *Internal, Parametric*) Knowledge
  - Knowledge that is difficult to articulate or express explicitly
  - Typically acquired through hands-on experience and practical application
    - e.g., Riding a Bicycle



Knowledge-Enhanced Reasoning

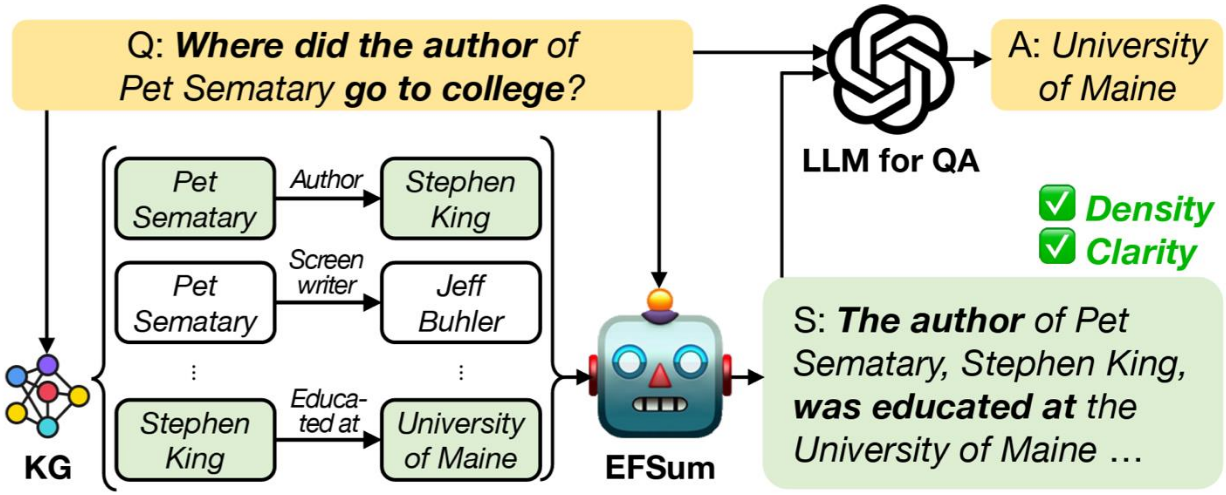
- Explicit Knowledge – Document Corpus
  - LLMs still have the tendency **to hallucinate** facts, **generate** inconsistent knowledge
  - **Retrieving prompts** for in-context learning is a nice means to achieve good performance
  - Persistently retrieving wiki documents for open-domain knowledge-intensive tasks that require complex multi-step reasoning



1) Trivedi et al., Interleaving Retrieval with Chain-of-Thought Reasoning for Knowledge-Intensive Multi-Step Questions, ACL 2023

Knowledge-Enhanced Reasoning

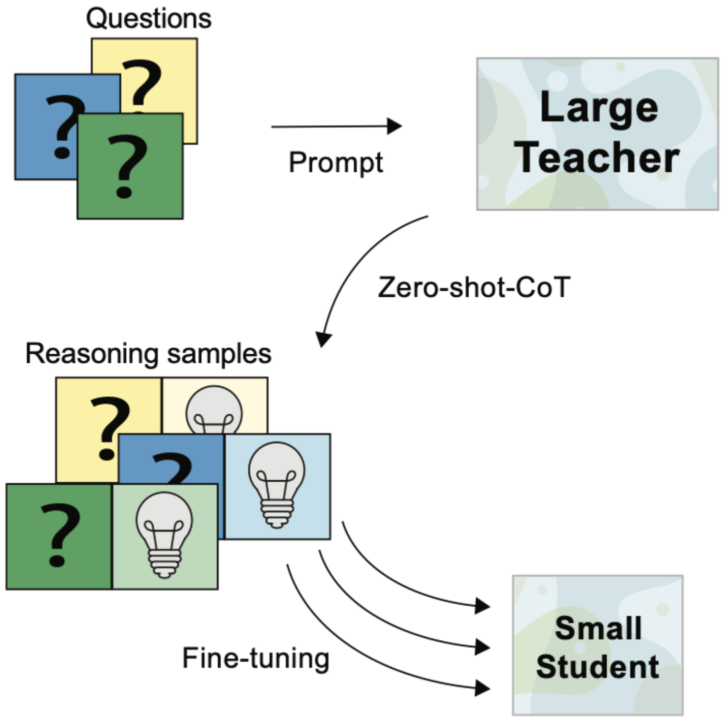
- Explicit Knowledge – Knowledge Graph (KGs)
  - Facts (or triples) from KGs are presented in a triple format: (*head*, *relation*, *tail*)
  - How can we effectively transform a set of relevant facts into a plausible text?
    - **Summarizing the facts** (density↑) while **highlighting the evidence** (clarity↑) to find the answer
  - How can we enhance **helpfulness** and **faithfulness** of the generated summaries?



1) Ko et al., Evidence-Focused Fact Summarization for Knowledge-Augmented Zero-Shot Question Answering, EMNLP 2024

Knowledge-Enhanced Reasoning

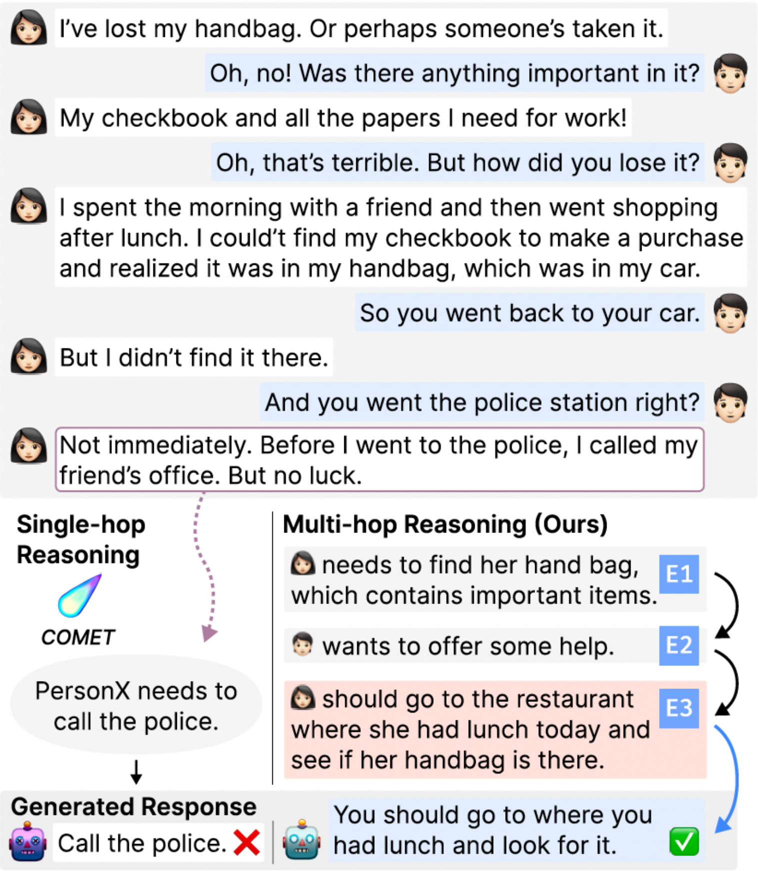
- Implicit Knowledge
  - LMs contain considerable **implicit knowledge**
  - Such “modeledge” can be induced as knowledge-informed prompts for reasoning
  - **Knowledge distillation** that generates reasoning samples by prompting a larger LM and teaches smaller LMs



1) Petroni et al., Language models as knowledge bases?, ACL 2019
2) Ho et al., Large Language Models Are Reasoning Teachers, ACL 2023

Reasoning Distillation for Conversation

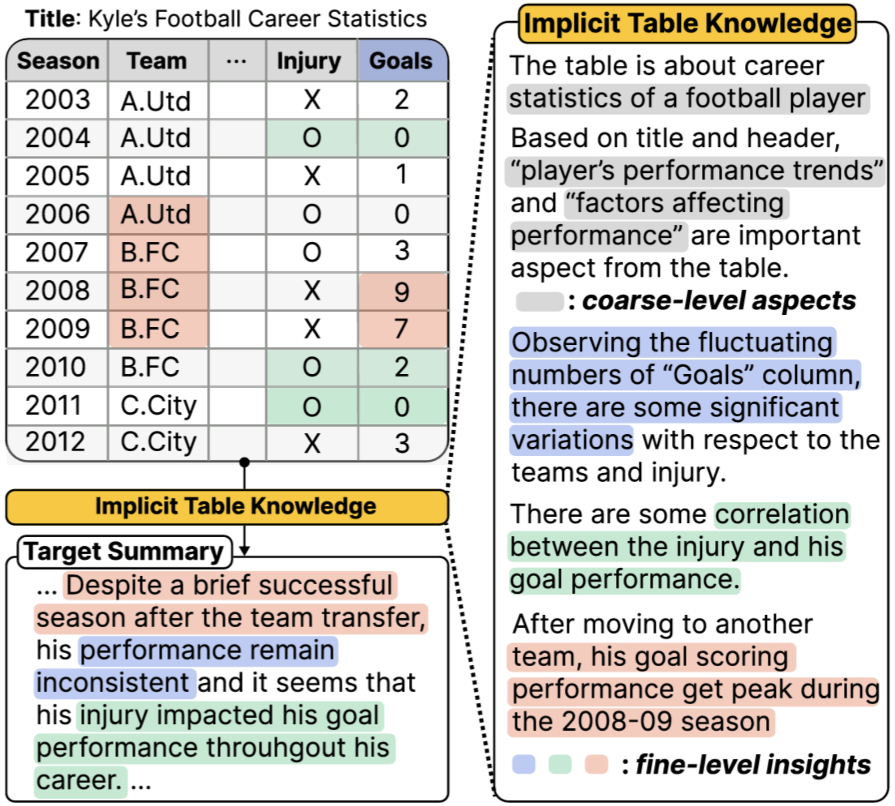
- Distillation for Commonsense Reasoning
  - Human-like chatbots necessitate the use of **commonsense reasoning**
    - It helps to comprehend and respond to implicit information present within conversations
  - LLMs generates **consistent and helpful rationales** and distills them into a small model
    - The small dialogue reasoner can generate CoT rationales by grounding on implicit evidence
    - The CoT rationales lead to high-quality responses



1) Chae et al., Dialogue Chain-of-Thought Distillation for Commonsense-aware Conversational Agents, EMNLP 2023

Reasoning Distillation for Data Analysis

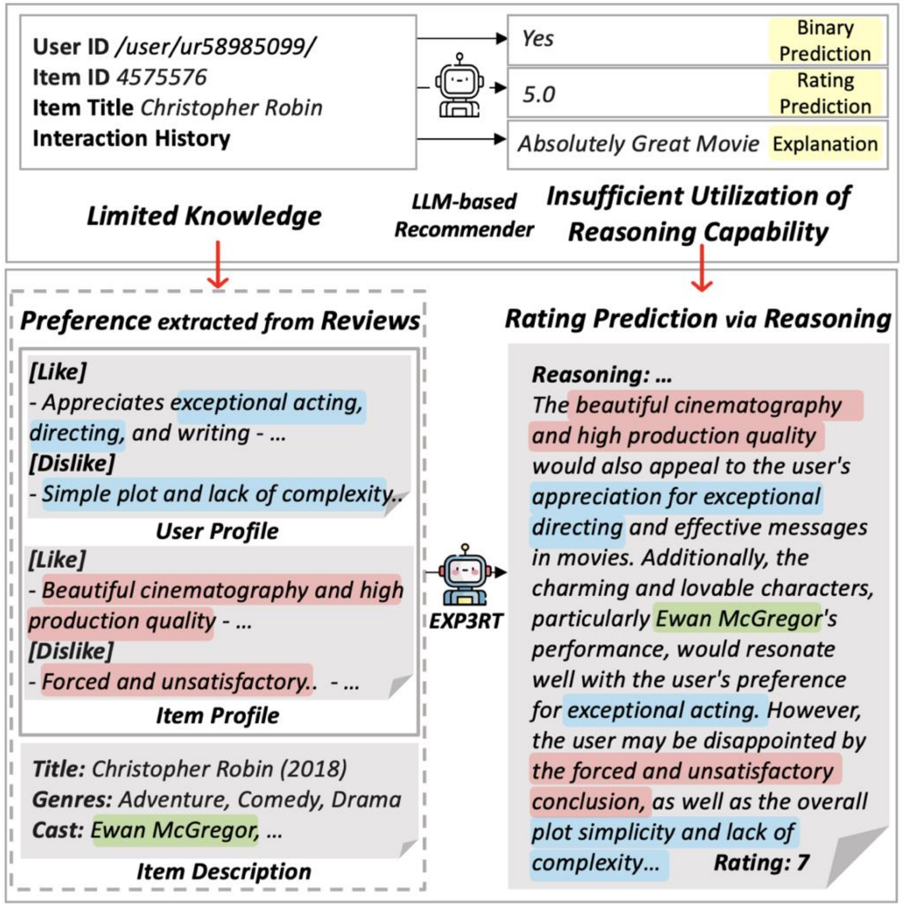
- Distillation for Table Reasoning
  - Generating **high-quality data insights** from tables requires uncovering implicit knowledge hidden within table cells
    - Mining implicit knowledge is challenging due to the complexity of structured tables
  - **Question-Then-Pinpoint (QtP)** self-questions insightful knowledge and pinpoints evidence directly from the table
    - This provides explainable guidance to support the summarization process



1) Seo et al., Unveiling Implicit Table Knowledge with Question-Then-Pinpoint Reasoner for Insightful Table Summarization, EMNLP Findings 2024

Reasoning Distillation for Recommendation

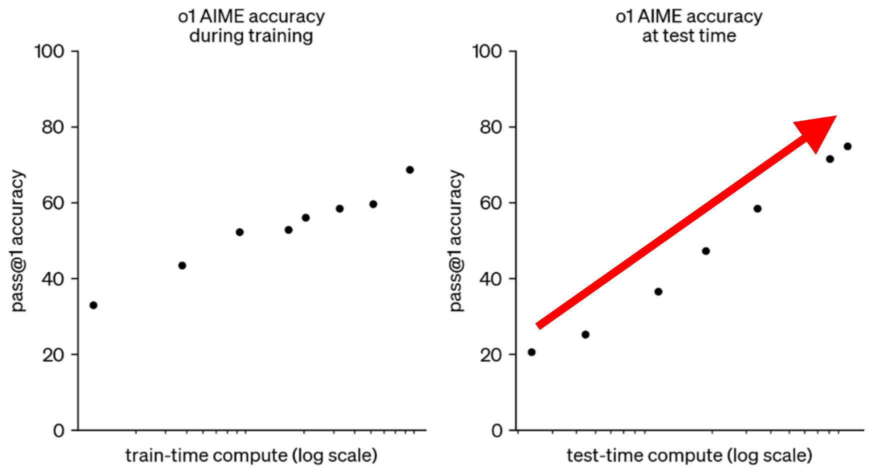
- Distillation for Preference Reasoning
  - Predicting a user’s rating score on an item requires personalized **preference reasoning**
    - Raw reviews of the user and item can provide valuable knowledge about their preference
    - The step-by-step reasoning for rating prediction provides logical and reasonable explanations
  - A student LLM is trained for **reasoning-enhanced rating prediction**
    - This delivers accurate predictions with transparent, human-readable explanations



1) Kim et al., Review-driven Personalized Preference Reasoning with Large Language Models for Recommendation, arXiv 2024

Train-Time vs. Test-Time Compute Scaling

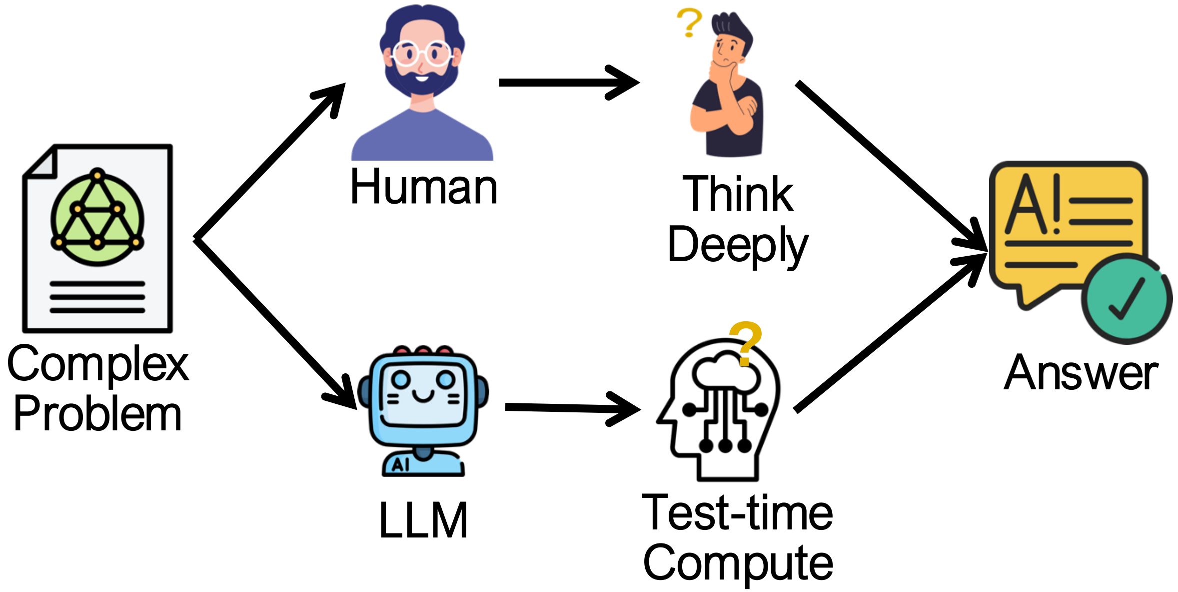
- **The scaling of train-time compute** has dominated the progress of LLMs
  - The resources needed to pretrain ever larger models are becoming **prohibitively expensive**, with billion-dollar clusters already on the horizon
- Rather than relying on ever-larger pretraining budgets, **test-time methods use dynamic inference strategies** that allow models to “think longer” on harder problems.



1) https://huggingface.co/spaces/HuggingFaceH4/blogpost-scaling-test-time-compute

Test-Time Compute Scaling

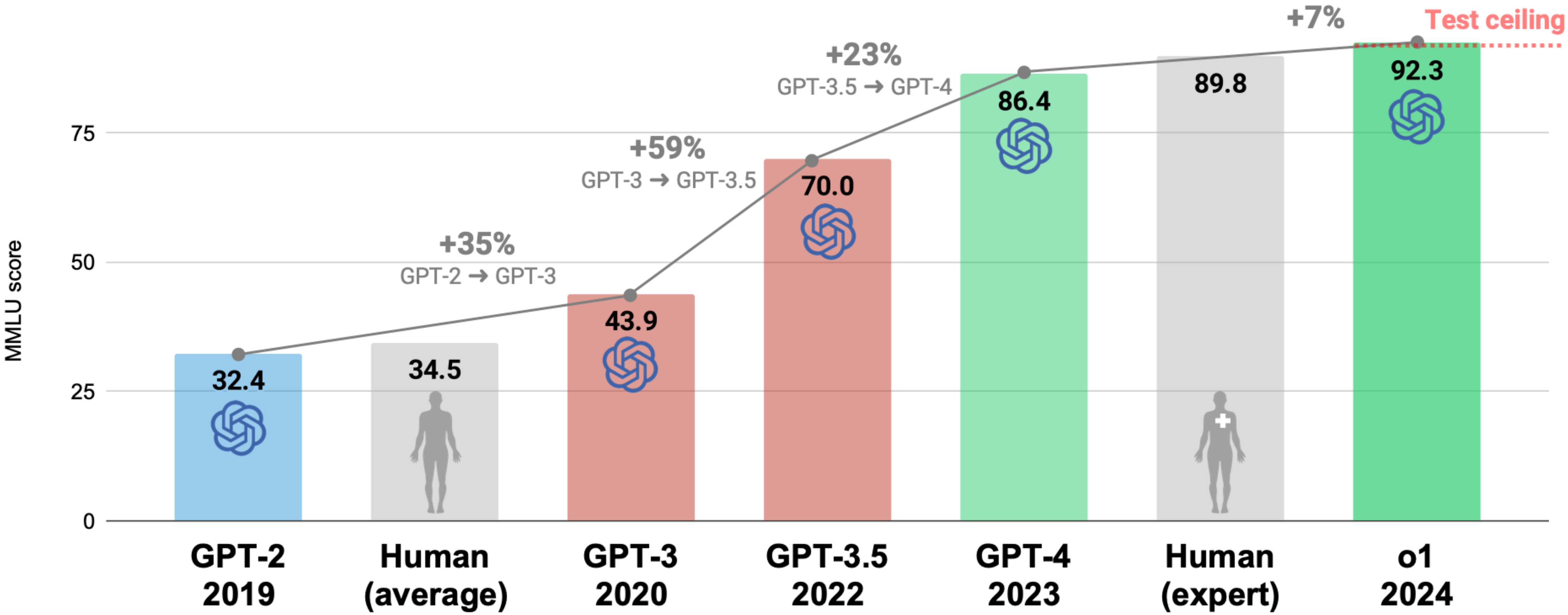
- When faced with a complex problem,
  - People **think deeply** through several stages before solving it.
  - LLMs can use **test-time compute** to improve their outputs



1) Snell et al., Scaling LLM Test-Time Compute Optimally can be More Effective than Scaling Model Parameters, Google DeepMind

Test-Time Compute Scaling: OpenAI's o1

- OpenAI's o1 model shows consistent improvement on difficult math problems as one increases the amount of test-time compute



MMLU (Massive Multitask Language Understanding) benchmark features 57 tasks including mathematics, US history, computer science, law, and more. % increases rounded. https://lifearchitect.ai/o1/ Alan D. Thompson. 2024.

Reward Modeling for "Correct Think"

- Process Reward Model (PRM)
  - PRM evaluates the reasoning process of a model and provides rewards
  - It improves the consistency and accuracy of inferences through step-by-step evaluation

correct

Let

$$x^8 + 3x^4 - 4 = p_1(x)p_2(x)\cdots p_k(x),$$

where each non-constant polynomial $p_i(x)$ is monic with integer coefficients, and cannot be factored further over the integers. Compute $p_1(1) + p_2(1) + \cdots + p_k(1)$.

I notice that the given polynomial has even degree and only even powers of x, so I can try to make a substitution to simplify it.

Let $y = x^4$, then the polynomial becomes $y^2 + 3y - 4$, which is a quadratic equation.

I can factor this quadratic equation as $(y+4)(y-1)$, so the original polynomial is $(x^4+4)(x^4-1)$.

Now I need to factor each of these factors further, if possible, over the integers.

For the first factor, $x^4 + 4$, I recall the Sophie Germain identity, which says that $a^4 + 4b^4 = (a^2 + 2b^2 + 2ab)(a^2 + 2b^2 - 2ab)$.

Applying this identity with $a = x$ and $b = 1$, I get $x^4 + 4 = (x^2 + 2x + 2)(x^2 - 2x + 2)$.

For the second factor, $x^4 - 1$, I can use the difference of squares formula, which says that $a^2 - b^2 = (a+b)(a-b)$.

Applying this formula with $a = x^2$ and $b = 1$, I get $x^4 - 1 = (x^2+1)(x^2-1)$.

Then I can apply the difference of squares formula again with $a = x$ and $b = 1$, to get $x^2 - 1 = (x+1)(x-1)$.

So the original polynomial is $(x^4+4)(x^4-1) = (x^2+2x+2)(x^2-2x+2)(x^2+1)(x+1)(x-1)$.

These are all the irreducible factors over the integers, so $p_1(x) = x^2+2x+2$, $p_2(x) = x^2-2x+2$, $p_3(x) = x^2+1$, $p_4(x) = x+1$, and $p_5(x) = x-1$.

To find the sum of their values at $x = 1$, I just need to plug in $x = 1$ into each factor and add them up.

I get $p_1(1) = 1^2 + 2\cdot 1 + 2 = 5$, $p_2(1) = 1^2 - 2\cdot 1 + 2 = 1$, $p_3(1) = 1^2 + 1 = 2$, $p_4(1) = 1 + 1 = 2$, and $p_5(1) = 1 - 1 = 0$.

So the final answer is $5 + 1 + 2 + 2 + 0 = 10$.

Answer: 10

incorrect

Let

$$x^8 + 3x^4 - 4 = p_1(x)p_2(x)\cdots p_k(x),$$

where each non-constant polynomial $p_i(x)$ is monic with integer coefficients, and cannot be factored further over the integers. Compute $p_1(1) + p_2(1) + \cdots + p_k(1)$.

This problem is about factoring a polynomial over the integers, which can be a tricky task.

One possible strategy is to look for rational roots of the polynomial, using the rational root theorem.

However, that would involve checking a lot of possible factors of -4, and there is no guarantee that the polynomial has any rational roots at all.

Another possible strategy is to use some algebraic tricks to simplify the polynomial or reduce its degree.

I notice that the polynomial has only even powers of x, so I can try to substitute $y = x^2$ and see what happens.

Then the polynomial becomes $y^4 + 3y^2 - 4$, which is a quadratic equation in y^2.

I can try to factor this quadratic equation by looking for two numbers that multiply to -4 and add to 3.

Those numbers are 4 and -1, so I can write $y^4 + 3y^2 - 4 = (y^2+4)(y^2-1)$.

Now I can substitute back $x^2 = y$ and get $x^8 + 3x^4 - 4 = (x^4+4)(x^4-1)$.

This looks promising, because I have reduced the degree of the factors from 8 to 4.

I can try to factor each of these factors further, using the difference of squares formula.

I get $x^8 + 3x^4 - 4 = (x^2+2)(x^2-2)(x^2+1)(x^2-1)$.

I can apply the difference of squares formula again to the last factor and get $x^8 + 3x^4 - 4 = (x^2+2)(x^2-2)(x^2+1)(x+1)(x-1)$.

Now I have factored the polynomial completely into monic linear and quadratic factors with integer coefficients.

These are the $p_i(x)$'s that the problem is asking for.

To find the sum of their values at $x = 1$, I just need to plug in $x = 1$ into each factor and add them up.

I get $p_1(1) + p_2(1) + \cdots + p_k(1) = (1^2+2)(1^2-2)(1^2+1)(1+1)(1-1)$.

Simplifying, I get $p_1(1) + p_2(1) + \cdots + p_k(1) = (3)(-1)(2)(2)(0)$.

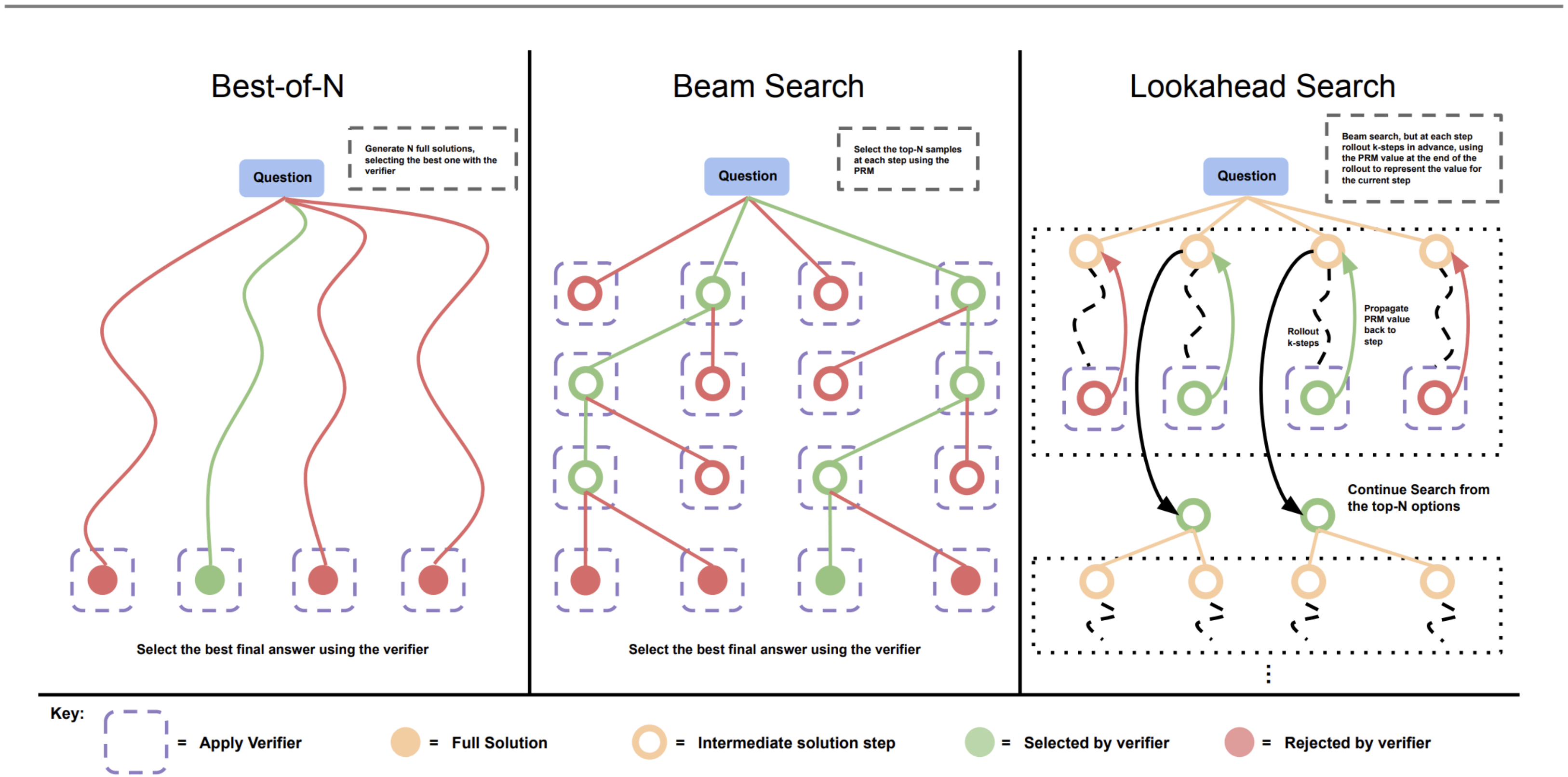
Multiplying, I get $p_1(1) + p_2(1) + \cdots + p_k(1) = 0$.

Answer: 0

1) Uesato, Jonathan, et al., Solving math word problems with process and outcome-based feedback, Google Deepmind 2022
2) Lightman et al., Let's Verify Step by Step, ICLR 2024

Strategies for Test-Time Compute Scaling

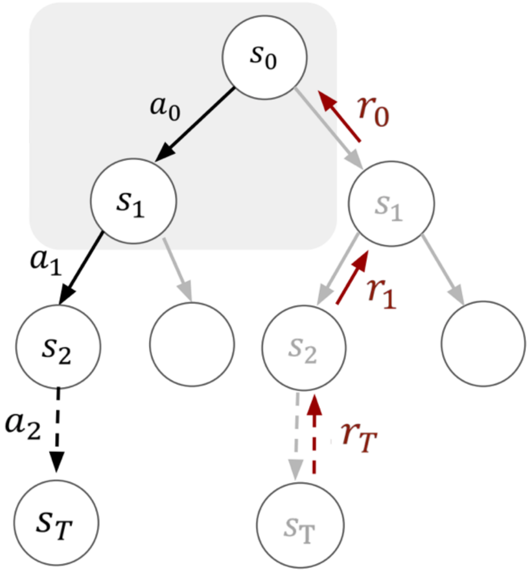
- Best-of-N vs. Beam Search vs. Lookahead Search

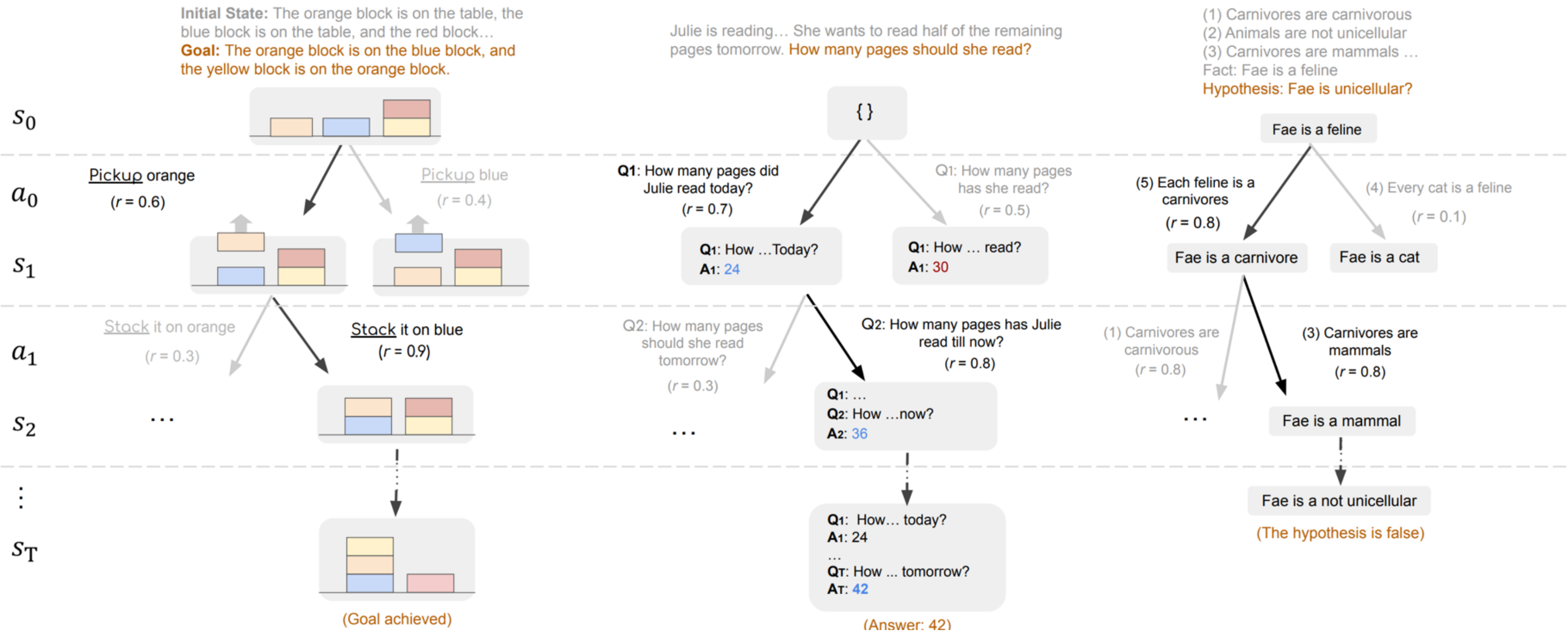


1) Snell et al., Scaling LLM Test-Time Compute Optimally can be More Effective than Scaling Model Parameters, Google DeepMind

Strategies for Test-Time Compute Scaling

- Lookahead Search – Monte Carlo Tree Search (MCTS)
  - MCTS optimizes search results through four steps for approaching the optimal solution
  - 1) selection, 2) expansion, 3) simulation, 4) backpropagation





1) Hao et al., Reasoning with Language Model is Planning with World Model, EMNLP 2023
2) Luo et al., Improve Mathematical Reasoning in Language Models by Automated Process Supervision, Google DeepMind 2024

Questions?